

What is the central idea in Atomic structure?

9701-01-C1

QUESTION

Which mechanism, condition or distinction matters in Atomic structure?

9701-01-C2

QUESTION

What is the electron configuration of Mg^{2+} ?

- A $1s^2 2s^2 2p^6$
 B $1s^2 2s^2 2p^6 3s^2$
 C $1s^2 2s^2 2p^5$
 D $1s^2 2s^2 2p^6 3s^1$

9701-01-M

QUESTION

State electron configuration of Al^{3+} ($Z = 13$) and explain why its third ionisation energy is lower than its fourth.

9701-01-P

QUESTION

What is the central idea in Atoms, molecules and stoichiometry?

9701-02-C1

QUESTION

Which mechanism, condition or distinction matters in Atoms, molecules and stoichiometry?

9701-02-C2

QUESTION

Which reagent limits when 1 mol N_2 reacts with 1 mol H_2 to make NH_3 ?

- A Nitrogen
 B Hydrogen
 C Both equal
 D Neither

9701-02-M

QUESTION

0.20 mol N_2 reacts with 0.45 mol H_2 : $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. Find limiting reagent and theoretical NH_3 amount.

9701-02-P

QUESTION

What is the central idea in Chemical bonding?

9701-03-C1

QUESTION

Which mechanism, condition or distinction matters in Chemical bonding?

9701-03-C2

QUESTION

Why is water polar?

- A It is linear
 B O-H bonds are nonpolar
 C Bent shape prevents dipole cancellation
 D It has no lone pairs

9701-03-M

QUESTION

Explain why water has a bent shape and a permanent dipole.

9701-03-P

QUESTION

What is the central idea in States of matter?

9701-04-C1

QUESTION

Which mechanism, condition or distinction matters in States of matter?

9701-04-C2

QUESTION

Which temperature must be used in $pV=nRT$?

- A Degrees Celsius
 B Degrees Fahrenheit
 C Any unconverted scale
 D Kelvin

9701-04-M

QUESTION

A. $1s^22s^22p^6$

Why: Atomic number counts protons; nucleon number counts protons and neutrons.

9701-01-M

ANSWER

Electron configurations fill orbitals according to energy, Pauli exclusion and Hund's rule. Successive ionisation energy jumps reveal an inner shell after removal of valence electrons.

9701-01-C2

ANSWER

Atomic number counts protons; nucleon number counts protons and neutrons. Isotopes have the same proton number but different neutron number.

9701-01-C1

ANSWER

A limiting reagent runs out first; compare available amount divided by stoichiometric coefficient. Percentage yield compares actual and theoretical product amounts.

9701-02-C2

ANSWER

Use $n = m/M$, $n = cV$ with V in dm^3 , and gas molar volume only under specified conditions. Balance equations before mole ratios.

9701-02-C1

ANSWER

Al^{3+} : $1s^22s^22p^6$. First three electrons include valence $3s^23p^1$; fourth is from inner $n = 2$ shell and is held more strongly.

9701-01-P

ANSWER

Ionic bonding is electrostatic attraction between oppositely charged ions. Covalent bonds share electron pairs; electronegativity difference gives polarity, but molecular shape determines overall dipole.

9701-03-C1

ANSWER

0.20 mol N_2 needs 0.60 mol H_2 , so H_2 limits. $\text{NH}_3 = (2/3)(0.45) = 0.30$ mol.

9701-02-P

ANSWER

B. Hydrogen
Why: Use $n = m/M$, $n = cV$ with V in dm^3 , and gas molar volume only under specified conditions.

9701-02-M

ANSWER

O has two bonding and two lone pairs; lone-pair repulsion produces bent arrangement; O-H bond dipoles do not cancel.

9701-03-P

ANSWER

C. Bent shape prevents dipole cancellation
Why: Ionic bonding is electrostatic attraction between oppositely charged ions.

9701-03-M

ANSWER

VSEPR predicts electron-pair arrangement; lone pairs repel more strongly than bonding pairs. Hydrogen bonding requires H bonded to N, O or F and a suitable lone pair.

9701-03-C2

ANSWER

D. Kelvin
Why: Ideal gas particles occupy negligible volume and have no intermolecular attraction.

9701-04-M

ANSWER

Solids, liquids and gases differ in particle arrangement, movement and proximity, while intermolecular forces influence melting and boiling.

9701-04-C2

ANSWER

Ideal gas particles occupy negligible volume and have no intermolecular attraction. $pV = nRT$ uses SI units and kelvin. Real gases deviate most at high pressure or low temperature where particle volume and attraction matter.

9701-04-C1

ANSWER

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4 States of matter

Calculate pressure of 0.050 mol ideal gas at 300 K in 1.0 dm³ (R = 8.31).

9701-04-P

QUESTION

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5 Chemical energetics

What is the central idea in Chemical energetics?

9701-05-C1

QUESTION

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5 Chemical energetics

Which mechanism, condition or distinction matters in Chemical energetics?

9701-05-C2

QUESTION

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5 Chemical energetics

Which sign represents an exothermic ΔH ?

- A Negative
- B Positive always
- C Zero
- D Undefined

9701-05-M

QUESTION

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5 Chemical energetics

A 0.050 mol reaction warms 200 g water by 5.0 K. Estimate molar ΔH using $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$.

9701-05-P

QUESTION

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6 Electrochemistry

What is the central idea in Electrochemistry?

9701-06-C1

QUESTION

FagusED · A-Level 9701
6 Electrochemistry

Which mechanism, condition or distinction matters in Electrochemistry?

9701-06-C2

QUESTION

FagusED · A-Level 9701
6 Electrochemistry

Which statement identifies oxidation?

- A Gain of protons
- B Loss of electrons
- C Loss of neutrons
- D Gain of electrons

9701-06-M

QUESTION

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6 Electrochemistry

Given $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.77 \text{ V}$ and $E^\circ(\text{I}_2/\text{I}^-) = +0.54 \text{ V}$, find E°_{cell} and identify the species oxidised.

9701-06-P

QUESTION

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7 Equilibria

What is the central idea in Equilibria?

9701-07-C1

QUESTION

FagusED · A-Level 9701
7 Equilibria

Which mechanism, condition or distinction matters in Equilibria?

9701-07-C2

QUESTION

FagusED · A-Level 9701
7 Equilibria

What changes K_c for a fixed reaction?

- A Adding more solid
- B Pressure at fixed T
- C Temperature
- D A catalyst at fixed T

9701-07-M

QUESTION

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7 Equilibria

For $\text{A} + \text{B} \rightleftharpoons 2\text{C}$, equilibrium concentrations are $[\text{A}] = 0.20$, $[\text{B}] = 0.50$, $[\text{C}] = 0.40 \text{ mol dm}^{-3}$. Calculate K_c .

9701-07-P

QUESTION

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8 Reaction kinetics

What is the central idea in Reaction kinetics?

9701-08-C1

QUESTION

FagusED · A-Level 9701
8 Reaction kinetics

Which mechanism, condition or distinction matters in Reaction kinetics?

9701-08-C2

QUESTION

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5 Chemical energetics

Bond energies approximate $\Delta H =$ bonds broken – bonds formed; a calorimeter uses $q = mc\Delta T$. Quote signs and account for losses when evaluating experimental values.

9701-05-C2

ANSWER

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5 Chemical energetics

Exothermic reactions have negative enthalpy change; endothermic positive. Hess's law lets enthalpy changes be added because enthalpy is a state function.

9701-05-C1

ANSWER

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4 States of matter

$V = 0.0010 \text{ m}^3$; $p = nRT/V = 1.25 \times 10^5 \text{ Pa}$.

9701-04-P

ANSWER

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6 Electrochemistry

Oxidation is electron loss, reduction electron gain. Oxidation numbers help balance redox equations, but charge and atoms must also balance.

9701-06-C1

ANSWER

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5 Chemical energetics

$q = 200 \times 4.18 \times 5 = 4180 \text{ J}$; $\Delta H = -4180/0.050 = -83\,600 \text{ J mol}^{-1} = -83.6 \text{ kJ mol}^{-1}$.

9701-05-P

ANSWER

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5 Chemical energetics

A. Negative
Why: Exothermic reactions have negative enthalpy change; endothermic positive.

9701-05-M

ANSWER

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6 Electrochemistry

Fe^{3+} reduced; I^- oxidised. $E^\circ_{\text{cell}} = 0.77 - 0.54 = +0.23 \text{ V}$.

9701-06-P

ANSWER

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6 Electrochemistry

B. Loss of electrons
Why: Oxidation is electron loss, reduction electron gain.

9701-06-M

ANSWER

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6 Electrochemistry

Standard electrode potentials describe reduction half-cells under standard conditions; positive $E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$ indicates thermodynamic feasibility under standard conditions, not reaction rate.

9701-06-C2

ANSWER

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7 Equilibria

C. Temperature
Why: A dynamic equilibrium has equal forward and reverse rates in a closed system, though concentrations need not be equal.

9701-07-M

ANSWER

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7 Equilibria

Changing temperature changes K_c ; changing concentration or pressure shifts composition while K_c stays fixed at fixed temperature.

9701-07-C2

ANSWER

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7 Equilibria

A dynamic equilibrium has equal forward and reverse rates in a closed system, though concentrations need not be equal. K_c uses equilibrium concentrations raised to stoichiometric powers; omit pure solids/liquids.

9701-07-C1

ANSWER

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8 Reaction kinetics

Collision theory requires sufficient energy and suitable orientation; increasing temperature raises the fraction with energy above activation energy. Catalysts offer a lower activation energy pathway and do not alter equilibrium position or K .

9701-08-C2

ANSWER

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8 Reaction kinetics

Rate can be measured from the tangent to concentration-time curves, gas collection or mass loss.

9701-08-C1

ANSWER

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7 Equilibria

$K_c = [\text{C}]^2/([\text{A}][\text{B}]) = 0.40^2/(0.20 \times 0.50) = 1.6$.

9701-07-P

ANSWER

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8 Reaction kinetics

What is the effect of a catalyst on activation energy?
A Raises it
B Eliminates collisions
C Changes ΔH
D Lowers it

9701-08-M

QUESTION

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8 Reaction kinetics

Explain why a catalyst increases both forward and reverse reaction rates but does not change K_c .

9701-08-P

QUESTION

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9 Periodicity

What is the central idea in Periodicity?

9701-09-C1

QUESTION

FagusED · A-Level 9701
9 Periodicity

Which mechanism, condition or distinction matters in Periodicity?

9701-09-C2

QUESTION

FagusED · A-Level 9701
9 Periodicity

Which oxide is generally more acidic?
A SO_3
B Na_2O
C MgO
D Al_2O_3 only

9701-09-M

QUESTION

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9 Periodicity

Explain why first ionisation energy of S is lower than P despite greater nuclear charge.

9701-09-P

QUESTION

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10 Group 2

What is the central idea in Group 2?

9701-10-C1

QUESTION

FagusED · A-Level 9701
10 Group 2

Which mechanism, condition or distinction matters in Group 2?

9701-10-C2

QUESTION

FagusED · A-Level 9701
10 Group 2

How does Group 2 hydroxide solubility generally change down the group?
A Decreases
B Increases
C Stays zero
D Cannot be measured

9701-10-M

QUESTION

FagusED · A-Level 9701
10 Group 2

Explain why $MgCO_3$ decomposes more readily than $BaCO_3$ on heating.

9701-10-P

QUESTION

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11 Group 17

What is the central idea in Group 17?

9701-11-C1

QUESTION

FagusED · A-Level 9701
11 Group 17

Which mechanism, condition or distinction matters in Group 17?

9701-11-C2

QUESTION

FagusED · A-Level 9701
11 Group 17

Which halogen oxidises Br^- to Br_2 in aqueous solution?
A I_2
B Br_2
C Cl_2
D Br^-

9701-11-M

QUESTION

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11 Group 17

Predict observation when chlorine water is added to aqueous potassium iodide, and write ionic equation.

9701-11-P

QUESTION

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12 Nitrogen and sulfur

What is the central idea in Nitrogen and sulfur?

9701-12-C1

QUESTION

Across Period 3 nuclear charge increases and atomic radius generally decreases as shielding changes little; first ionisation energy broadly increases with subshell/pairing exceptions.

9701-09-C1

ANSWER

It provides lower activation-energy route for both directions; equilibrium composition/thermodynamics at fixed T are unchanged.

9701-08-P

ANSWER

D. Lowers it
Why: Rate can be measured from the tangent to concentration-time curves, gas collection or mass loss.

9701-08-M

ANSWER

In sulfur one 3p orbital has paired electrons; inter-electron repulsion makes one easier to remove than the unpaired 3p electron in phosphorus.

9701-09-P

ANSWER

A. SO_3
Why: Across Period 3 nuclear charge increases and atomic radius generally decreases as shielding changes little; first ionisation energy broadly increases with subshell/pairing exceptions.

9701-09-M

ANSWER

Oxides change from basic metallic through amphoteric to acidic non-metallic. Explain anomalies with orbital energy and electron repulsion, not a blanket memorised trend.

9701-09-C2

ANSWER

B. Increases
Why: Down Group 2, ionisation energy decreases due to increased distance and shielding.

9701-10-M

ANSWER

Hydroxide solubility rises down the group; sulfate solubility falls. Thermal decomposition of carbonates/nitrates becomes harder as larger cations polarise anions less strongly.

9701-10-C2

ANSWER

Down Group 2, ionisation energy decreases due to increased distance and shielding. Metal reactivity with water increases.

9701-10-C1

ANSWER

Halide ions give silver halide precipitates with AgNO_3 after acidification; colour and ammonia solubility help identify ions.

9701-11-C2

ANSWER

Halogen oxidising strength decreases down the group; halide reducing ability increases. A more reactive halogen displaces a less reactive halide from solution.

9701-11-C1

ANSWER

Mg^{2+} is smaller and more strongly polarises carbonate ion, weakening C-O bonding and lowering thermal stability.

9701-10-P

ANSWER

Ammonia's lone pair makes it a base and ligand. The Haber process balances equilibrium yield with rate and cost using moderate conditions and a catalyst; catalyst does not shift equilibrium.

9701-12-C1

ANSWER

Brown iodine appears; $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$.

9701-11-P

ANSWER

C. Cl_2
Why: Halogen oxidising strength decreases down the group; halide reducing ability increases.

9701-11-M

ANSWER

FagusED · A-Level 9701
12 Nitrogen and sulfur

Which mechanism, condition or distinction matters in Nitrogen and sulfur?

9701-12-C2

QUESTION

FagusED · A-Level 9701
12 Nitrogen and sulfur

What does an industrial catalyst change in Haber equilibrium?
A Position of equilibrium
B K_c
C Enthalpy change
D Rate of attainment

9701-12-M

QUESTION

FagusED · A-Level 9701
12 Nitrogen and sulfur

Explain why industrial ammonia production uses a catalyst and recycles unreacted gases.

9701-12-P

QUESTION

FagusED · A-Level 9701
13 Introduction to organic chemistry

What is the central idea in Introduction to organic chemistry?

9701-13-C1

QUESTION

FagusED · A-Level 9701
13 Introduction to organic chemistry

Which mechanism, condition or distinction matters in Introduction to organic chemistry?

9701-13-C2

QUESTION

FagusED · A-Level 9701
13 Introduction to organic chemistry

Which pair can be E/Z isomers?
A But-2-ene
B Ethene
C Propene
D 2-methylpropene

9701-13-M

QUESTION

FagusED · A-Level 9701
13 Introduction to organic chemistry

Explain why $\text{CH}_3\text{CH}=\text{CHCH}_3$ can have E/Z isomers but $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ cannot.

9701-13-P

QUESTION

FagusED · A-Level 9701
14 Hydrocarbons

What is the central idea in Hydrocarbons?

9701-14-C1

QUESTION

FagusED · A-Level 9701
14 Hydrocarbons

Which mechanism, condition or distinction matters in Hydrocarbons?

9701-14-C2

QUESTION

FagusED · A-Level 9701
14 Hydrocarbons

Which reaction class describes HBr addition to propene?
A Free-radical substitution
B Electrophilic addition
C Elimination
D Condensation

9701-14-M

QUESTION

FagusED · A-Level 9701
14 Hydrocarbons

State the mechanism for adding HBr to propene and predict major product using carbocation stability.

9701-14-P

QUESTION

FagusED · A-Level 9701
15 Halogen compounds

What is the central idea in Halogen compounds?

9701-15-C1

QUESTION

FagusED · A-Level 9701
15 Halogen compounds

Which mechanism, condition or distinction matters in Halogen compounds?

9701-15-C2

QUESTION

FagusED · A-Level 9701
15 Halogen compounds

Hot ethanolic OH^- typically promotes which haloalkane reaction?
A Oxidation
B Polymerisation
C Elimination
D Ester hydrolysis

9701-15-M

QUESTION

FagusED · A-Level 9701
15 Halogen compounds

Predict organic product when 2-bromopropane reacts with hot ethanolic KOH and explain the role of base.

9701-15-P

QUESTION

FagusED · A-Level 9701
12 Nitrogen and sulfur

Catalyst speeds forward and reverse reactions; does not alter equilibrium yield; recycling unreacted N_2 and H_2 raises overall conversion.

9701-12-P

ANSWER

FagusED · A-Level 9701
12 Nitrogen and sulfur

D. Rate of attainment
Why: Ammonia's lone pair makes it a base and ligand.

9701-12-M

ANSWER

FagusED · A-Level 9701
12 Nitrogen and sulfur

Sulfur dioxide is an acidic, reducing oxide and an atmospheric pollutant; sulfuric acid manufacture involves reversible oxidation to SO_3 .

9701-12-C2

ANSWER

FagusED · A-Level 9701
13 Introduction to organic chemistry

A. But-2-ene
Why: Functional groups predict characteristic reactions.

9701-13-M

ANSWER

FagusED · A-Level 9701
13 Introduction to organic chemistry

Curly arrows show movement of an electron pair from its source to an electron-deficient site; distinguish electrophile, nucleophile and free radical.

9701-13-C2

ANSWER

FagusED · A-Level 9701
13 Introduction to organic chemistry

Functional groups predict characteristic reactions. Structural isomers have the same molecular formula but different connectivity; stereoisomers have the same connectivity but different spatial arrangement.

9701-13-C1

ANSWER

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14 Hydrocarbons

Benzene is stabilised by delocalised π electrons and typically substitutes rather than adds, preserving aromaticity.

9701-14-C2

ANSWER

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14 Hydrocarbons

Alkanes undergo free-radical substitution with halogens under UV light, often producing mixtures. Alkenes undergo electrophilic addition because π electrons attract electrophiles.

9701-14-C1

ANSWER

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13 Introduction to organic chemistry

Each alkene carbon in but-2-ene has two different groups; terminal carbon of but-1-ene has two H atoms.

9701-13-P

ANSWER

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15 Halogen compounds

Haloalkanes undergo nucleophilic substitution when a nucleophile attacks δ^+ carbon; aqueous hydroxide produces an alcohol, while ethanolic hydroxide favours elimination to alkene.

9701-15-C1

ANSWER

FagusED · A-Level 9701
14 Hydrocarbons

Electrophilic addition; H^+ adds to terminal carbon to form more stable secondary carbocation; major product 2-bromopropane.

9701-14-P

ANSWER

FagusED · A-Level 9701
14 Hydrocarbons

B. Electrophilic addition
Why: Alkanes undergo free-radical substitution with halogens under UV light, often producing mixtures.

9701-14-M

ANSWER

FagusED · A-Level 9701
15 Halogen compounds

Propene; OH^- removes β -hydrogen, and HBr is eliminated to make $\text{C}=\text{C}$.

9701-15-P

ANSWER

FagusED · A-Level 9701
15 Halogen compounds

C. Elimination
Why: Haloalkanes undergo nucleophilic substitution when a nucleophile attacks δ^+ carbon; aqueous hydroxide produces an alcohol, while ethanolic hydroxide favours elimination to alkene.

9701-15-M

ANSWER

FagusED · A-Level 9701
15 Halogen compounds

Rate depends on bond strength and structure. Halogenoarenes resist substitution under mild conditions due to resonance affecting C-halogen bond.

9701-15-C2

ANSWER

FagusED · A-Level 9701
16 Hydroxy compounds

What is the central idea in Hydroxy compounds?

9701-16-C1

QUESTION

FagusED · A-Level 9701
16 Hydroxy compounds

Which mechanism, condition or distinction matters in Hydroxy compounds?

9701-16-C2

QUESTION

FagusED · A-Level 9701
16 Hydroxy compounds

Which alcohol typically oxidises to a ketone?
A Primary
B Tertiary
C Phenol only
D Secondary

9701-16-M

QUESTION

FagusED · A-Level 9701
16 Hydroxy compounds

Outline conditions and product for oxidising ethanol to ethanoic acid rather than ethanal.

9701-16-P

QUESTION

FagusED · A-Level 9701
17 Carbonyl compounds

What is the central idea in Carbonyl compounds?

9701-17-C1

QUESTION

FagusED · A-Level 9701
17 Carbonyl compounds

Which mechanism, condition or distinction matters in Carbonyl compounds?

9701-17-C2

QUESTION

FagusED · A-Level 9701
17 Carbonyl compounds

Which carbonyl compound forms a silver mirror with Tollens' reagent?
A Aldehyde
B Ketone only
C Alkane
D Ester only

9701-17-M

QUESTION

FagusED · A-Level 9701
17 Carbonyl compounds

An unknown C_3H_6O compound forms a silver mirror and is reduced to propan-1-ol. Identify it and justify.

9701-17-P

QUESTION

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

What is the central idea in Carboxylic acids and derivatives?

9701-18-C1

QUESTION

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

Which mechanism, condition or distinction matters in Carboxylic acids and derivatives?

9701-18-C2

QUESTION

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

What is produced when ethanol reacts with ethanoic acid under suitable conditions?
A Ethanamide
B Ethyl ethanoate
C Ethanoyl chloride
D Ethene

9701-18-M

QUESTION

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

Name ester from propanoic acid and methanol, and state two conditions/features of preparation.

9701-18-P

QUESTION

FagusED · A-Level 9701
19 Nitrogen compounds

What is the central idea in Nitrogen compounds?

9701-19-C1

QUESTION

FagusED · A-Level 9701
19 Nitrogen compounds

Which mechanism, condition or distinction matters in Nitrogen compounds?

9701-19-C2

QUESTION

FagusED · A-Level 9701
19 Nitrogen compounds

What acts as a Brønsted base in amines?
A N-H bond only
B Carbon skeleton
C Nitrogen lone pair
D Carboxyl oxygen always

9701-19-M

QUESTION

FagusED · A-Level 9701
16 Hydroxy compounds

D. Secondary
Why: Alcohol oxidation depends on carbon bearing -OH: primary gives aldehyde then acid; secondary gives ketone; tertiary resists mild oxidation.

9701-16-M

ANSWER

FagusED · A-Level 9701
16 Hydroxy compounds

Conditions matter: distil aldehyde to prevent further oxidation, reflux for acid. Phenol is more acidic than an alcohol because phenoxide ion is resonance stabilised.

9701-16-C2

ANSWER

FagusED · A-Level 9701
16 Hydroxy compounds

Alcohol oxidation depends on carbon bearing -OH: primary gives aldehyde then acid; secondary gives ketone; tertiary resists mild oxidation.

9701-16-C1

ANSWER

FagusED · A-Level 9701
17 Carbonyl compounds

NaBH₄ reduces aldehydes to primary alcohols and ketones to secondary alcohols. HCN addition creates cyanohydrins and may create a chiral centre.

9701-17-C2

ANSWER

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17 Carbonyl compounds

Aldehydes and ketones have polar C=O bonds susceptible to nucleophilic addition. Aldehydes oxidise with Tollens' or Fehling's reagents while ketones generally do not.

9701-17-C1

ANSWER

FagusED · A-Level 9701
16 Hydroxy compounds

Use excess acidified dichromate(VI); heat under reflux; product ethanoic acid.

9701-16-P

ANSWER

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

Carboxylic acids donate H⁺ because carboxylate is resonance stabilised. Esters form by acid-catalysed condensation of acid and alcohol; hydrolysis reverses ester formation.

9701-18-C1

ANSWER

FagusED · A-Level 9701
17 Carbonyl compounds

Propanal; aldehyde reduces Tollens' reagent; carbonyl reduction gives primary alcohol.

9701-17-P

ANSWER

FagusED · A-Level 9701
17 Carbonyl compounds

A. Aldehyde
Why: Aldehydes and ketones have polar C=O bonds susceptible to nucleophilic addition.

9701-17-M

ANSWER

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

Methyl propanoate; acid catalyst e.g. concentrated H₂SO₄; heat under reflux/reversible condensation with water.

9701-18-P

ANSWER

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

B. Ethyl ethanoate
Why: Carboxylic acids donate H⁺ because carboxylate is resonance stabilised.

9701-18-M

ANSWER

FagusED · A-Level 9701
18 Carboxylic acids and derivatives

Acyl chlorides are more reactive than esters and undergo nucleophilic acyl substitution, giving HCl with alcohols or water.

9701-18-C2

ANSWER

FagusED · A-Level 9701
19 Nitrogen compounds

C. Nitrogen lone pair
Why: Amines are bases because the nitrogen lone pair accepts H⁺; electron donation by alkyl groups can alter basicity.

9701-19-M

ANSWER

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19 Nitrogen compounds

Amines react with acids to form ammonium salts and with acyl chlorides to form amides. Amino acids have both amino and carboxyl groups and often exist as zwitterions depending on pH.

9701-19-C2

ANSWER

FagusED · A-Level 9701
19 Nitrogen compounds

Amines are bases because the nitrogen lone pair accepts H⁺; electron donation by alkyl groups can alter basicity.

9701-19-C1

ANSWER

FagusED · A-Level 9701
19 Nitrogen compounds

Explain why glycine can migrate toward a cathode at sufficiently low pH and show its predominant charged form.

9701-19-P

QUESTION

FagusED · A-Level 9701
20 Polymerisation

What is the central idea in Polymerisation?

9701-20-C1

QUESTION

FagusED · A-Level 9701
20 Polymerisation

Which mechanism, condition or distinction matters in Polymerisation?

9701-20-C2

QUESTION

FagusED · A-Level 9701
20 Polymerisation

What small molecule is often lost during polyester formation?

- A O₂
- B N₂
- C Br₂
- D H₂O

9701-20-M

QUESTION

FagusED · A-Level 9701
20 Polymerisation

Identify polymerisation type for ethane-1,2-diol and ethanedioic acid, and name the linkage and small molecule released.

9701-20-P

QUESTION

FagusED · A-Level 9701
21 Organic synthesis

What is the central idea in Organic synthesis?

9701-21-C1

QUESTION

FagusED · A-Level 9701
21 Organic synthesis

Which mechanism, condition or distinction matters in Organic synthesis?

9701-21-C2

QUESTION

FagusED · A-Level 9701
21 Organic synthesis

How is overall yield of successive steps found?

- A Multiply fractional yields
- B Add percentage yields
- C Use only the first step
- D Divide yield by step count

9701-21-M

QUESTION

FagusED · A-Level 9701
21 Organic synthesis

A synthesis has yields of 70%, 80% and 50% for three steps. Calculate overall percentage yield and give one reason isolated yield may be low.

9701-21-P

QUESTION

FagusED · A-Level 9701
22 Analytical techniques

What is the central idea in Analytical techniques?

9701-22-C1

QUESTION

FagusED · A-Level 9701
22 Analytical techniques

Which mechanism, condition or distinction matters in Analytical techniques?

9701-22-C2

QUESTION

FagusED · A-Level 9701
22 Analytical techniques

Which IR feature suggests an O-H group?

- A Only a ¹³C NMR peak
- B Broad O-H absorption
- C A chlorine isotope pair
- D Absence of all absorption

9701-22-M

QUESTION

FagusED · A-Level 9701
22 Analytical techniques

A compound shows molecular ion m/z 46 and broad O-H IR absorption; ¹H NMR has signals integrating 3:2:1. Suggest a structure and explain integration.

9701-22-P

QUESTION

FagusED · A-Level 9701
23 Chemical energetics

What is the central idea in Chemical energetics?

9701-23-C1

QUESTION

FagusED · A-Level 9701
23 Chemical energetics

Which mechanism, condition or distinction matters in Chemical energetics?

9701-23-C2

QUESTION

FagusED · A-Level 9701
20 Polymerisation

Hydrolysable ester and amide links make some condensation polymers biodegradable; properties depend on structure, crystallinity and cross-linking.

9701-20-C2

ANSWER

FagusED · A-Level 9701
20 Polymerisation

Addition polymerisation joins alkene monomers without small-molecule by-products. Condensation polymerisation links bifunctional monomers and removes small molecules, commonly water or HCl.

9701-20-C1

ANSWER

FagusED · A-Level 9701
19 Nitrogen compounds

At low pH amino group is protonated and carboxyl group mainly COOH ; $\text{H}_3\text{N}^+\text{CH}_2\text{COOH}$ has net + charge and moves toward cathode.

9701-19-P

ANSWER

FagusED · A-Level 9701
21 Organic synthesis

A multi-step synthesis needs a feasible path between functional groups, with reagents and conditions chosen to control competing reactions.

9701-21-C1

ANSWER

FagusED · A-Level 9701
20 Polymerisation

Condensation; ester link; water.

9701-20-P

ANSWER

FagusED · A-Level 9701
20 Polymerisation

D. H_2O
Why: Addition polymerisation joins alkene monomers without small-molecule by-products.

9701-20-M

ANSWER

FagusED · A-Level 9701
21 Organic synthesis

$0.70 \times 0.80 \times 0.50 = 0.28$, or 28%; side reactions, incomplete conversion or purification losses.

9701-21-P

ANSWER

FagusED · A-Level 9701
21 Organic synthesis

A. Multiply fractional yields
Why: A multi-step synthesis needs a feasible path between functional groups, with reagents and conditions chosen to control competing reactions.

9701-21-M

ANSWER

FagusED · A-Level 9701
21 Organic synthesis

Overall yield is the product of fractional step yields; purification may reduce isolated yield but increase purity. Check carbon count and oxidation state at every step.

9701-21-C2

ANSWER

FagusED · A-Level 9701
22 Analytical techniques

B. Broad O-H absorption
Why: Infrared absorption helps identify functional groups; a broad O-H band differs for alcohol and carboxylic acid.

9701-22-M

ANSWER

FagusED · A-Level 9701
22 Analytical techniques

Mass spectra provide molecular-ion mass and fragmentation clues; isotope patterns can indicate halogens. ^1H NMR signals reflect chemically distinct proton environments, with integration showing relative counts and splitting from neighbouring protons.

9701-22-C2

ANSWER

FagusED · A-Level 9701
22 Analytical techniques

Infrared absorption helps identify functional groups; a broad O-H band differs for alcohol and carboxylic acid.

9701-22-C1

ANSWER

FagusED · A-Level 9701
23 Chemical energetics

A negative ΔG does not ensure fast reaction. Standard enthalpy changes may be derived from formation data using products minus reactants.

9701-23-C2

ANSWER

FagusED · A-Level 9701
23 Chemical energetics

At A Level, entropy measures dispersal of energy and arrangements; $\Delta G = \Delta H - T\Delta S$ predicts feasible direction at specified temperature under standard conditions.

9701-23-C1

ANSWER

FagusED · A-Level 9701
22 Analytical techniques

Ethanol $\text{CH}_3\text{CH}_2\text{OH}$; CH_3 gives 3 H, CH_2 gives 2 H, OH gives 1 H.

9701-22-P

ANSWER

What criterion of ΔG supports feasibility at specified conditions?
 A $\Delta G > 0$ always
 B $\Delta G = \Delta H$ only
 C $\Delta G < 0$
 D ΔG depends only on catalyst

9701-23-M

QUESTION

A process has $\Delta H = +75 \text{ kJ mol}^{-1}$ and $\Delta S = +0.150 \text{ kJ mol}^{-1} \text{ K}^{-1}$. Find threshold temperature for $\Delta G < 0$.

9701-23-P

QUESTION

What is the central idea in Electrochemistry?

9701-24-C1

QUESTION

Which mechanism, condition or distinction matters in Electrochemistry?

9701-24-C2

QUESTION

How many moles electrons deposit 1 mol Cu from Cu^{2+} ?
 A 0.5
 B 1
 C 3
 D 2

9701-24-M

QUESTION

Copper is deposited by $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ using 1.0 A for 1930 s. Find amount of Cu formed.

9701-24-P

QUESTION

What is the central idea in Equilibria?

9701-25-C1

QUESTION

Which mechanism, condition or distinction matters in Equilibria?

9701-25-C2

QUESTION

For equal $[\text{HA}]$ and $[\text{A}^-]$ in an acid buffer, pH approximately equals what?
 A pK_a
 B K_w
 C 0
 D 14 always

9701-25-M

QUESTION

An acid buffer has $\text{pK}_a = 4.76$ and $[\text{A}^-]/[\text{HA}] = 10$. Determine pH and explain effect of adding a little HCl.

9701-25-P

QUESTION

What is the central idea in Reaction kinetics?

9701-26-C1

QUESTION

Which mechanism, condition or distinction matters in Reaction kinetics?

9701-26-C2

QUESTION

When doubling $[\text{A}]$ quadruples rate, order in A is what?
 A First
 B Second
 C Zero
 D Fourth

9701-26-M

QUESTION

When $[\text{A}]$ doubles at fixed $[\text{B}]$, rate rises fourfold; when $[\text{B}]$ doubles, rate is unchanged. Write rate law and overall order.

9701-26-P

QUESTION

What is the central idea in Group 2?

9701-27-C1

QUESTION

FagusED · A-Level 9701
24 Electrochemistry

The Nernst equation accounts for concentrations in electrode potential; E changes with reaction quotient rather than just the standard value.

9701-24-C1

ANSWER

FagusED · A-Level 9701
23 Chemical energetics

Set $\Delta G = 0$, $T = \Delta H / \Delta S = 75 / 0.150 = 500 \text{ K}$; feasible by ΔG criterion above 500 K.

9701-23-P

ANSWER

FagusED · A-Level 9701
23 Chemical energetics

C. $\Delta G < 0$
Why: At A Level, entropy measures dispersal of energy and arrangements; $\Delta G = \Delta H - T\Delta S$ predicts feasible direction at specified temperature under standard conditions.

9701-23-M

ANSWER

FagusED · A-Level 9701
24 Electrochemistry

$Q = It = 1930 \text{ C}$; electrons = $1930 / 96500 = 0.020 \text{ mol}$; Cu = 0.010 mol .

9701-24-P

ANSWER

FagusED · A-Level 9701
24 Electrochemistry

D. 2
Why: The Nernst equation accounts for concentrations in electrode potential; E changes with reaction quotient rather than just the standard value.

9701-24-M

ANSWER

FagusED · A-Level 9701
24 Electrochemistry

For a spontaneous cell reaction $\Delta G = -nFE$, with n the electrons transferred. Electrolysis requires charge $Q = It$ and amount electrons Q/F ; electron stoichiometry determines product amount.

9701-24-C2

ANSWER

FagusED · A-Level 9701
25 Equilibria

A. pKa
Why: Weak acid dissociation uses $K_a = [H^+][A^-] / [HA]$, while $K_w = [H^+][OH^-]$.

9701-25-M

ANSWER

FagusED · A-Level 9701
25 Equilibria

For acid buffer, $pH = pK_a + \log([A^-] / [HA])$ when conditions justify approximation.

9701-25-C2

ANSWER

FagusED · A-Level 9701
25 Equilibria

Weak acid dissociation uses $K_a = [H^+][A^-] / [HA]$, while $K_w = [H^+][OH^-]$. A buffer contains a weak acid and conjugate base: added H^+ is consumed by A^- and added OH^- by HA.

9701-25-C1

ANSWER

FagusED · A-Level 9701
26 Reaction kinetics

A first-order reaction has constant half-life at fixed temperature. The Arrhenius equation links temperature, activation energy and rate constant; a catalyst lowers activation energy.

9701-26-C2

ANSWER

FagusED · A-Level 9701
26 Reaction kinetics

A rate law is determined experimentally: rate = $k[A]^m[B]^n$. Reaction order is not inferred automatically from balanced stoichiometry.

9701-26-C1

ANSWER

FagusED · A-Level 9701
25 Equilibria

$pH = 4.76 + \log_{10}(10) = 5.76$; A^- consumes much of added H^+ , reducing pH change.

9701-25-P

ANSWER

FagusED · A-Level 9701
27 Group 2

For Group 2 compounds, ionic size and charge density explain solubility and thermal stability trends; hydration and lattice enthalpies compete in dissolving salts.

9701-27-C1

ANSWER

FagusED · A-Level 9701
26 Reaction kinetics

Order 2 in A, order 0 in B; rate = $k[A]^2$, overall order 2.

9701-26-P

ANSWER

FagusED · A-Level 9701
26 Reaction kinetics

B. Second
Why: A rate law is determined experimentally: rate = $k[A]^m[B]^n$.

9701-26-M

ANSWER

FagusED · A-Level 9701
27 Group 2

Which mechanism, condition or distinction matters in Group 2?

9701-27-C2

QUESTION

FagusED · A-Level 9701
27 Group 2

Which compound forms a white insoluble precipitate in the sulfate test?

- A $\text{Mg}(\text{OH})_2$
- B CaO
- C BaSO_4
- D $\text{Ba}(\text{OH})_2$

9701-27-M

QUESTION

FagusED · A-Level 9701
27 Group 2

Predict solubility trend of $\text{Mg}(\text{OH})_2$ to $\text{Ba}(\text{OH})_2$ and explain why sulfate tests use barium salts.

9701-27-P

QUESTION

FagusED · A-Level 9701
28 Transition elements

What is the central idea in Transition elements?

9701-28-C1

QUESTION

FagusED · A-Level 9701
28 Transition elements

Which mechanism, condition or distinction matters in Transition elements?

9701-28-C2

QUESTION

FagusED · A-Level 9701
28 Transition elements

Which ion has incomplete d subshell?

- A Zn^{2+}
- B Sc^{3+}
- C Cu^+
- D Cu^{2+}

9701-28-M

QUESTION

FagusED · A-Level 9701
28 Transition elements

Explain why Zn^{2+} is usually colourless and $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ coloured.

9701-28-P

QUESTION

FagusED · A-Level 9701
29 Introduction to organic chemistry

What is the central idea in Introduction to organic chemistry?

9701-29-C1

QUESTION

FagusED · A-Level 9701
29 Introduction to organic chemistry

Which mechanism, condition or distinction matters in Introduction to organic chemistry?

9701-29-C2

QUESTION

FagusED · A-Level 9701
29 Introduction to organic chemistry

Why is a racemate optically inactive?
A Equal opposite rotations cancel
B No chiral molecules exist
C Molecules are all planar
D It cannot absorb light

9701-29-M

QUESTION

FagusED · A-Level 9701
29 Introduction to organic chemistry

Why is a racemic mixture optically inactive even if each pure enantiomer rotates plane-polarised light?

9701-29-P

QUESTION

FagusED · A-Level 9701
30 Hydrocarbons

What is the central idea in Hydrocarbons?

9701-30-C1

QUESTION

FagusED · A-Level 9701
30 Hydrocarbons

Which mechanism, condition or distinction matters in Hydrocarbons?

9701-30-C2

QUESTION

FagusED · A-Level 9701
30 Hydrocarbons

What is the electrophile in benzene nitration?

- A NO_3^-
- B NO_2^+
- C NH_4^+
- D OH^-

9701-30-M

QUESTION

FagusED · A-Level 9701
30 Hydrocarbons

Name electrophile in benzene nitration and explain why substitution predominates over addition.

9701-30-P

QUESTION

FagusED · A-Level 9701
27 Group 2

Hydroxide solubility increases down Group 2; BaSO_4 has very low solubility, forming a visible white precipitate with sulfate ions.

9701-27-P

ANSWER

FagusED · A-Level 9701
27 Group 2

C. BaSO_4
Why: For Group 2 compounds, ionic size and charge density explain solubility and thermal stability trends; hydration and lattice enthalpies compete in dissolving salts.

9701-27-M

ANSWER

FagusED · A-Level 9701
27 Group 2

Group 2 hydroxides become more soluble down the group; sulfate solubility decreases. Industrial/analytical uses depend on insolubility and acid-base behaviour.

9701-27-C2

ANSWER

FagusED · A-Level 9701
28 Transition elements

D. Cu^{2+}
Why: Transition elements have atoms or stable ions with incomplete d subshells.

9701-28-M

ANSWER

FagusED · A-Level 9701
28 Transition elements

Ligands donate electron pairs to a central ion; ligand substitution can change colour and coordination geometry. Colour depends on splitting and absorption of visible light; d^0/d^{10} ions are often colourless.

9701-28-C2

ANSWER

FagusED · A-Level 9701
28 Transition elements

Transition elements have atoms or stable ions with incomplete d subshells. Variable oxidation states, coloured complexes and catalysis arise from d-electron behaviour.

9701-28-C1

ANSWER

FagusED · A-Level 9701
29 Introduction to organic chemistry

Optical isomerism occurs when a tetrahedral carbon has four different groups; a racemate has equal enantiomers and no net optical rotation. Mechanisms must account for electron movement and the stability of intermediates.

9701-29-C2

ANSWER

FagusED · A-Level 9701
29 Introduction to organic chemistry

At A Level, organic pathways integrate reaction mechanisms, stereochemistry and aromaticity.

9701-29-C1

ANSWER

FagusED · A-Level 9701
28 Transition elements

Zn^{2+} has full $3d^{10}$ so no d-d transitions; Cu^{2+} has incomplete 3d subshell; ligand-field splitting allows visible photon absorption.

9701-28-P

ANSWER

FagusED · A-Level 9701
30 Hydrocarbons

Aromatic reactions commonly proceed by electrophilic substitution to preserve benzene's delocalised π system.

9701-30-C1

ANSWER

FagusED · A-Level 9701
29 Introduction to organic chemistry

The mixture has equal amounts of opposite enantiomers; rotations are equal and opposite so cancel.

9701-29-P

ANSWER

FagusED · A-Level 9701
29 Introduction to organic chemistry

A. Equal opposite rotations cancel
Why: At A Level, organic pathways integrate reaction mechanisms, stereochemistry and aromaticity.

9701-29-M

ANSWER

FagusED · A-Level 9701
30 Hydrocarbons

NO_2^+ ; addition would destroy the stable delocalised aromatic system, while substitution restores it after deprotonation.

9701-30-P

ANSWER

FagusED · A-Level 9701
30 Hydrocarbons

B. NO_2^+
Why: Aromatic reactions commonly proceed by electrophilic substitution to preserve benzene's delocalised π system.

9701-30-M

ANSWER

FagusED · A-Level 9701
30 Hydrocarbons

Nitration uses nitronium ion generated by concentrated nitric and sulfuric acids; Friedel-Crafts reactions use a catalyst. Electron-donating/withdrawing substituents alter ring reactivity and direct substitution position.

9701-30-C2

ANSWER

FagusED · A-Level 9701
31 Halogen compounds

What is the central idea in Halogen compounds?

9701-31-C1

QUESTION

FagusED · A-Level 9701
31 Halogen compounds

Which mechanism, condition or distinction matters in Halogen compounds?

9701-31-C2

QUESTION

FagusED · A-Level 9701
31 Halogen compounds

Which conditions promote haloalkane elimination?
A Cold aqueous OH⁻
B Aqueous AgNO₃
C Hot ethanolic OH⁻
D Dry HCl only

9701-31-M

QUESTION

FagusED · A-Level 9701
31 Halogen compounds

Describe two changes to favour elimination rather than substitution of 2-bromobutane and give two possible alkene products.

9701-31-P

QUESTION

FagusED · A-Level 9701
32 Hydroxy compounds

What is the central idea in Hydroxy compounds?

9701-32-C1

QUESTION

FagusED · A-Level 9701
32 Hydroxy compounds

Which mechanism, condition or distinction matters in Hydroxy compounds?

9701-32-C2

QUESTION

FagusED · A-Level 9701
32 Hydroxy compounds

Which reagent reacts with phenol but typically not ethanol in acid-base test?
A Water
B Sodium metal
C Bromine water only
D Aqueous NaOH

9701-32-M

QUESTION

FagusED · A-Level 9701
32 Hydroxy compounds

Phenol and ethanol both react with Na metal. Give one reagent that distinguishes them and expected result.

9701-32-P

QUESTION

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

What is the central idea in Carboxylic acids and derivatives?

9701-33-C1

QUESTION

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

Which mechanism, condition or distinction matters in Carboxylic acids and derivatives?

9701-33-C2

QUESTION

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

What is a by-product of ethanoyl chloride reacting with ethanol?
A HCl
B H₂
C O₂
D NH₃

9701-33-M

QUESTION

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

Name product from ethanoyl chloride and ethanol and describe the reaction type and another product.

9701-33-P

QUESTION

FagusED · A-Level 9701
34 Nitrogen compounds

What is the central idea in Nitrogen compounds?

9701-34-C1

QUESTION

FagusED · A-Level 9701
34 Nitrogen compounds

Which mechanism, condition or distinction matters in Nitrogen compounds?

9701-34-C2

QUESTION

FagusED · A-Level 9701
34 Nitrogen compounds

Which is more basic in water?
A Ethanamide
B Ethanamine
C Both equal by definition
D Neither can protonate

9701-34-M

QUESTION

FagusED · A-Level 9701
31 Halogen compounds

C. Hot ethanolic OH⁻
Why: Aryl halides are less reactive in nucleophilic substitution than haloalkanes because carbon-halogen bonding to the aromatic ring is strengthened by resonance and the sp² carbon environment.

9701-31-M

ANSWER

FagusED · A-Level 9701
31 Halogen compounds

Haloalkane elimination competes with substitution; solvent, temperature and substrate structure influence outcome.

9701-31-C2

ANSWER

FagusED · A-Level 9701
31 Halogen compounds

Aryl halides are less reactive in nucleophilic substitution than haloalkanes because carbon-halogen bonding to the aromatic ring is strengthened by resonance and the sp² carbon environment.

9701-31-C1

ANSWER

FagusED · A-Level 9701
32 Hydroxy compounds

Comparing reactions with Na, NaOH and Na₂CO₃ distinguishes acidity levels: typical phenol reacts with NaOH but not sodium carbonate under usual conditions.

9701-32-C2

ANSWER

FagusED · A-Level 9701
32 Hydroxy compounds

Phenol is more acidic than aliphatic alcohol because the conjugate phenoxide ion delocalises negative charge. Phenol activates its ring toward bromination and nitration.

9701-32-C1

ANSWER

FagusED · A-Level 9701
31 Halogen compounds

Use ethanolic rather than aqueous base and heat; but-1-ene and but-2-ene.

9701-31-P

ANSWER

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

Acyl chlorides, acid anhydrides, esters and amides vary in ease of nucleophilic acyl substitution. Nucleophile attacks carbonyl C, tetrahedral intermediate forms and leaving group departs.

9701-33-C1

ANSWER

FagusED · A-Level 9701
32 Hydroxy compounds

Aqueous NaOH; phenol dissolves/reacts forming phenoxide; ethanol shows no comparable acid-base reaction.

9701-32-P

ANSWER

FagusED · A-Level 9701
32 Hydroxy compounds

D. Aqueous NaOH
Why: Phenol is more acidic than aliphatic alcohol because the conjugate phenoxide ion delocalises negative charge.

9701-32-M

ANSWER

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

Ethyl ethanoate; nucleophilic acyl substitution/esterification; HCl.

9701-33-P

ANSWER

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

A. HCl
Why: Acyl chlorides, acid anhydrides, esters and amides vary in ease of nucleophilic acyl substitution.

9701-33-M

ANSWER

FagusED · A-Level 9701
33 Carboxylic acids and derivatives

Amides are resonance stabilised and less reactive; hydrolysis may require acid/base and heat. Amino acids condense to peptide bonds.

9701-33-C2

ANSWER

FagusED · A-Level 9701
34 Nitrogen compounds

B. Ethanamine
Why: Amines act as bases and nucleophiles; amides are much less basic because the nitrogen lone pair delocalises into the carbonyl.

9701-34-M

ANSWER

FagusED · A-Level 9701
34 Nitrogen compounds

Amino acids exist as zwitterions near their isoelectric point, affecting solubility and electrophoretic movement. Diazonium compounds can form azo dyes via coupling with activated aromatic rings under suitable cold conditions.

9701-34-C2

ANSWER

FagusED · A-Level 9701
34 Nitrogen compounds

Amines act as bases and nucleophiles; amides are much less basic because the nitrogen lone pair delocalises into the carbonyl.

9701-34-C1

ANSWER

FagusED · A-Level 9701
34 Nitrogen compounds

Compare basicity of ethanamine and ethanamide using their nitrogen lone pairs.

9701-34-P

QUESTION

FagusED · A-Level 9701
35 Polymerisation

What is the central idea in Polymerisation?

9701-35-C1

QUESTION

FagusED · A-Level 9701
35 Polymerisation

Which mechanism, condition or distinction matters in Polymerisation?

9701-35-C2

QUESTION

FagusED · A-Level 9701
35 Polymerisation

Which polymer backbone lacks hydrolysable ester or amide links?
A Polyester
B Polyamide
C Poly(ethene)
D Protein

9701-35-M

QUESTION

FagusED · A-Level 9701
35 Polymerisation

Explain why a polyamide may be hydrolysed by aqueous acid but poly(ethene) is relatively resistant.

9701-35-P

QUESTION

FagusED · A-Level 9701
36 Organic synthesis

What is the central idea in Organic synthesis?

9701-36-C1

QUESTION

FagusED · A-Level 9701
36 Organic synthesis

Which mechanism, condition or distinction matters in Organic synthesis?

9701-36-C2

QUESTION

FagusED · A-Level 9701
36 Organic synthesis

What is a necessary check in retrosynthetic route planning?
A Ignore carbon count
B Only estimate colour
C Never select conditions
D Account for functional-group changes

9701-36-M

QUESTION

FagusED · A-Level 9701
36 Organic synthesis

A three-step synthesis gives 80% yield at each step. Calculate overall yield and compare with a one-step 50% yield.

9701-36-P

QUESTION

FagusED · A-Level 9701
37 Analytical techniques

What is the central idea in Analytical techniques?

9701-37-C1

QUESTION

FagusED · A-Level 9701
37 Analytical techniques

Which mechanism, condition or distinction matters in Analytical techniques?

9701-37-C2

QUESTION

FagusED · A-Level 9701
37 Analytical techniques

An M and M+2 mass peak in near 3:1 ratio suggests which atom?
A Chlorine
B Carbon only
C Nitrogen
D Hydrogen

9701-37-M

QUESTION

FagusED · A-Level 9701
35 Polymerisation

A repeat unit should show bonds that extend to adjacent units, and monomers can be deduced by cutting linkage bonds and restoring their functional groups.

9701-35-C2

ANSWER

FagusED · A-Level 9701
35 Polymerisation

Condensation polyamides and polyesters are vulnerable to hydrolysis because their links can be cleaved; addition polymers with C-C backbones resist hydrolysis.

9701-35-C1

ANSWER

FagusED · A-Level 9701
34 Nitrogen compounds

Ethanamine is more basic; its N lone pair is available to accept H⁺; amide N lone pair delocalises into C=O and is less available.

9701-34-P

ANSWER

FagusED · A-Level 9701
36 Organic synthesis

Plan retrosynthetically from the target, identify the key bond/functional group, then reverse to feasible starting materials.

9701-36-C1

ANSWER

FagusED · A-Level 9701
35 Polymerisation

Amide C-N links undergo hydrolysis; acid catalyses cleavage; poly(ethene) C-C backbone lacks such hydrolysable functional groups.

9701-35-P

ANSWER

FagusED · A-Level 9701
35 Polymerisation

C. Poly(ethene)
Why: Condensation polyamides and polyesters are vulnerable to hydrolysis because their links can be cleaved; addition polymers with C-C backbones resist hydrolysis.

9701-35-M

ANSWER

FagusED · A-Level 9701
36 Organic synthesis

$0.8^3 = 0.512$ or 51.2%; three-step route has slightly greater overall yield than 50%.

9701-36-P

ANSWER

FagusED · A-Level 9701
36 Organic synthesis

D. Account for functional-group changes
Why: Plan retrosynthetically from the target, identify the key bond/functional group, then reverse to feasible starting materials.

9701-36-M

ANSWER

FagusED · A-Level 9701
36 Organic synthesis

Forward route needs reagents, solvents and conditions; include isolation and purification. Percentage yields compound across steps and can make short routes preferable even if individual reactions look efficient.

9701-36-C2

ANSWER

FagusED · A-Level 9701
37 Analytical techniques

A. Chlorine
Why: A Level spectra may combine high-resolution mass spectrometry, IR, ¹H/¹³C NMR and chromatography.

9701-37-M

ANSWER

FagusED · A-Level 9701
37 Analytical techniques

Chromatographic R_f = distance spot/distance solvent front for the stated stationary and mobile phases.

9701-37-C2

ANSWER

FagusED · A-Level 9701
37 Analytical techniques

A Level spectra may combine high-resolution mass spectrometry, IR, ¹H/¹³C NMR and chromatography. Use molecular formula/degree of unsaturation first, then identify carbonyl and O-H signals and proton environments; no single peak proves an entire structure.

9701-37-C1

ANSWER

FagusED · A-Level 9701
37 Analytical techniques

A mass spectrum suggests formula C_3H_6O and IR has strong $C=O$; 1H NMR has one proton environment. Suggest structure and justify.

9701-37-P

QUESTION

FagusED · A-Level 9701
P3 Advanced practical skills

What is the central idea in P3 Advanced practical skills?

9701-38-C1

QUESTION

FagusED · A-Level 9701
P3 Advanced practical skills

Which mechanism, condition or distinction matters in P3 Advanced practical skills?

9701-38-C2

QUESTION

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P3 Advanced practical skills

Which good titration practice improves reliability?
A One rough titre only
B Repeat to concordance
C Read volume to nearest litre
D Rinse burette with analyte

9701-38-M

QUESTION

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P3 Advanced practical skills

A mean 25.0 cm^3 titre of 0.100 mol dm^{-3} HCl neutralises 25.0 cm^3 NaOH. Determine NaOH concentration and state one titration good practice.

9701-38-P

QUESTION

FagusED · A-Level 9701
P5 Planning, analysis and evaluation

What is the central idea in P5 Planning, analysis and evaluation?

9701-39-C1

QUESTION

FagusED · A-Level 9701
P5 Planning, analysis and evaluation

Which mechanism, condition or distinction matters in P5 Planning, analysis and evaluation?

9701-39-C2

QUESTION

FagusED · A-Level 9701
P5 Planning, analysis and evaluation

For a fair rate comparison, what must be controlled?
A Two variables changed together
B Initial-rate units omitted
C Temperature
D Graph slope hidden

9701-39-M

QUESTION

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P5 Planning, analysis and evaluation

Outline a fair comparison of reaction rate at two acid concentrations using gas volume.

9701-39-P

QUESTION

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P3 Advanced practical skills

Qualitative tests need reagent, observation and inference, plus alternatives where observations overlap.

9701-38-C2

ANSWER

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P3 Advanced practical skills

Use safe measured quantities and observations to make reproducible chemical inferences. In titration rinse burette with titrant, pipette with analyte and repeat to concordance; record readings with appropriate decimal places.

9701-38-C1

ANSWER

FagusED · A-Level 9701
37 Analytical techniques

Propanone CH_3COCH_3 ; equivalent methyl groups give one ^1H environment; carbonyl accounts for IR and formula.

9701-37-P

ANSWER

FagusED · A-Level 9701
P5 Planning, analysis and evaluation

A plan identifies reaction, independent and dependent variables, controls and a measurable endpoint.

9701-39-C1

ANSWER

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P3 Advanced practical skills

$n(\text{HCl}) = 0.00250 \text{ mol}$; 1:1 gives $[\text{NaOH}] = 0.100 \text{ mol dm}^{-3}$; use concordant titres/read burette to two decimal places.

9701-38-P

ANSWER

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P3 Advanced practical skills

B. Repeat to concordance
Why: Use safe measured quantities and observations to make reproducible chemical inferences.

9701-38-M

ANSWER

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P5 Planning, analysis and evaluation

Keep acid volume, solid mass/area and temperature constant; change acid concentration; measure initial gradient of gas-volume against time with repeat trials.

9701-39-P

ANSWER

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P5 Planning, analysis and evaluation

C. Temperature
Why: A plan identifies reaction, independent and dependent variables, controls and a measurable endpoint.

9701-39-M

ANSWER

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P5 Planning, analysis and evaluation

Use a graph to distinguish orders or calculate an enthalpy with $q = mc\Delta T$, then evaluate heat losses and significant uncertainty. Safety controls should address actual reagents, not an unconnected generic warning.

9701-39-C2

ANSWER